MIL-S-6252D
19 NOVEMBER 1958
SUPERSEDING
MIL-I-6252C
31 AUGUST 1955

MILITARY SPECIFICATION

SPECIFICATIONS, DETAIL, FIXED WING AIRCRAFT, PREPARATION OF

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification contains instructions to be followed in the preparation of detail specifications for the complete aircraft. These instructions are to govern the detail specification writer and establish the format of the detail specification, but are not to be construed as requirements applicable to the design or construction of aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-D-5028 — Drawings and Data Lists; Preparation of Manufacturing (For Production, Aeronautical and Associated Equipment).

MIL-D-5480 — Data, Engineering and Technical (Reproduction Thereof).

3. REQUIREMENTS

3.1 The aircraft detail specification shall present, in summary form, all pertinent data relative to the design and, in addition, shall make reference to any other data which may be necessary for a complete description of the aircraft or design. Drawings and related data shall be prepared in accordance with Specification MIL-D-5028 or as specified by the procuring activity.

3.2 Descriptions shall be complete and precise. Detail specifications shall indicate precisely the intentions of the bidder and, later. his obligations as a contractor, the requirements which define the characteristics of the aircraft, the type of construction, details, et cetera, and the degree of compliance with requirements. An exact repetition of the language of those paragraphs in the requirements documents that are suggestive of the various arrangements, con-

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structions, et cetera, or cover desired requirements, is not satisfactory and shall be avoided. However, the foregoing does not preclude a repetition of requirement documents paragraphs that contain mandatory requirements. Any paragraph so repeated shall be expanded, if necessary, to give a concise yet complete description of the part, arrangement, construction proposed, et cetera. The assignment of part numbers and drawing numbers which describe the details of the aircraft shall be made in accordance with Specification MIL–D–5028.

3.3 Paragraph numbers shall be used in the order indicated in this specification with the addition of subparagraphs, sub-subparagraphs, et cetera, as required to complete the descriptions or to supply the information required hereinafter in the specification. (In some cases, additional instructions are contained in parentheses appearing in the various paragraphs and sections of this specification.) The subparagraph numbering shall follow the sequence indicated by the order of the items in the parenthetical instructions. Where (sub) paragraph headings in this specification are not applicable, but are headings which would affect the numbering sequence of succeeding (sub) paragraphs, the (sub) paragraph number and heading shall be listed with a notation "Not applicable". When the numbering sequence of subsequent applicable (sub) paragraphs will not be affected, the (sub) paragraph(s) may be deleted.

3.3.1 In the case of detail specifications for Navy aircraft, where a main topic or subtopic listed herein may be applicable to aircraft as a class, but not required for a specific aircraft, the heading shall be listed with the notation "Not required".

3.3.2 Unless otherwise specified by the procuring activity, detail specifications for follow-on or repeat production contracts shall be prepared using a paragraph sequence numbering the same as used for the detail specification on which the first or prior production contract was based. (This exception will prevent confusion to users of the detail specification for repeat production orders of the same model aircraft.)

3.4 Applicable to U. S. Air Force (USAF). In order to facilitate the preparation and checking of detail specifications, a number within a ¼-inch diameter circle, located approximately opposite the center of the paragraph affected, shall be placed in the left margin, opposite any paragraph in which a deviation from a governmental requirement is described or implied. This number shall correspond to the number assigned to the particular deviation as covered in appendix II. Failure to list deviations shall be construed as intent to comply. Deviation numbers in detail specifications for follow-on or repeat contracts shall remain the same as those assigned to the deviation in the original detail specification. The numbers of individual deviations which are not required for follow-on or repeat contracts shall not be assigned to different deviations, but shall be included in the proper sequence with the notation "Not applicable".

3.5 Contractor's model designation and specification.

3.5.1 *Contractor's model designation.* A separate and distinct contractor's model designation shall be given to each basic model aircraft. Modification of the basic model aircraft will be designated using a suffixed letter to the basic designation. A separate and distinct model designation shall be assigned to each proposed aircraft design which involves difference in performance guaranty, configuration, major constructional difference, engine, or engine-propeller combination:

Example 1:

Model 278 Specification 4028, Basic.

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Model 278a Specification 4029, Modification No. 1.

Model 278b Specification 4030, Modification No. 2.

Example 2:

Model 278 Specification 4028, Basic.

Model 278a Specification 4029, Modification No. 1 and Appendix IV (Optional arrangements not affecting performance guaranty).

3.5.2 *Contractor's detail specification number.* A separate and distinct specification number shall be given to each basic detail specification or modification thereof.

3.5.3 *Contractor's detail specification.* The detail specification shall be separate and complete for each contractor's basic model of aircraft offered. Sheets or appendixes indicating changes from a basic detail specification will not be acceptable as a detail specification of the model offered. In the event that optional arrangements not affecting the performance guaranties are offered, they may be submitted as appendixes to the applicable detail specification.

3.5.4 *Binding.* Each of the copies of the detail specification, together with all appendixes, shall be bound under one cover, preferably of the loose leaf binder style so that revision pages may be inserted. The final specification which is furnished at the end of the contract however, may be furnished with a different type of binder cover.

3.6 Air Force model designation. When the Air Force model designation of an aircraft is changed, such as C–40 to C–40A, a completely revised detail specification shall be furnished, and amendments or appendixes are not acceptable. The contractor shall assign a new number for each specification and a suffixed revision letter to the final corrected copy.

Example:

C–40 B7598 Final corrected copy C–40 B7598A.

C–40A B7600 Final corrected copy C–40A B7600A.

C–41 B7980 Final corrected copy C–41A B7980A.

Note: This applies to aircraft that have been procured on Air Force contracts, and to which Air Force model designations have been assigned.

3.7 Form of presentation. Unless otherwise specified, the detail specification shall be furnished in the format as indicated by this specification and having an approximate page size of 8 by 10½ inches. There shall be 1¼ inch margin on the left-hand side of each page. The detail specification shall be typewritten, using any standard typewriter. Capital letters shall not be used exclusively in the text.

3.8 Quantity. Unless otherwise specified by the procuring activity prior to approval of the detail specification, 25 prints, dark lettering on white background, such as blueline prints, shall be furnished. The approved detail specification shall be supplied to the procuring activity as one set of legible, direct reading reproducible prints, preferably van dykes, and 50 legible print or offset copies, unless quantities or distribution different from this is requested by the procuring activity, shall be furnished not later than 30 days after receipt of approval of the particular detail specification by the contractor. The reproducible copies shall meet the legibility requirements of Specification MIL–D–5480.

3.9 Changes to aircraft detail specification.

3.9.1 *Applicable to U. S. Air Force (USAF).* To facilitate keeping the aircraft detail specification up-to-date as changes are made, contractual Aircraft Specification Change Notice Sheets, prepared in accordance with appendix III, for filing opposite each page of the detail specification affected as approved by the procuring ac-

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tivity, shall be forwarded to the procuring activity in order to provide an up-to-date specification between complete specification revisions. Any new specifications shall be complete in themselves. Quantity of Change Notice Sheets shall be in accordance with 3.8.

3.9.1.2 *Final corrected detail specification.* The aircraft detail specification shall be corrected as a continuous reading specification which describes the last aircraft delivered on the contract. This specification shall be the basic contract specification for procurement of any subsequent articles of this model aircraft. The quantity of copies of the final corrected detail specification shall be as specified by the procuring activity.

3.9.2 *Applicable to Navy-Bureau of Aeronautics (BuAer).* At intervals of not more than 120 days, the contractor shall submit sets of detail specification revised pages and indexes suitable for insertion in detail specifications in loose leaf, or equivalent, binder. The revised pages shall be prepared in accordance with current BuAer instructions. The quantity and distribution of sets of revised pages shall be as specified by BuAer. The revised pages shall apply to the contract detail specification.

3.10 Government-furnished aircraft equipment (GFAE) lists. The lists of Government furnished aircraft equipment (GFAE) which appear in appendixes I–A and I–B reflect the initial equipment that the contractor plans to use or that the Government plans to furnish, as applicable. These lists will be revised to reflect authorized changes.

3.11 Format and content. The detail specification shall conform to the following:

i. COVER SHEET (The cover sheet shall contain information giving aircraft model designation, specification number, date, manufacturer's name, and contract or proposal number (or other identification).)

ii. CONTENTS (A table of contents giving the location of the main breakdowns shall be provided at this point, giving the numbers, titles, and page numbers of the main paragraphs and subparagraphs, as well as the page numbers of the beginnings of the various appendixes. An alphabetical index shall be included after the last appendix of the specification if specified by the procuring activity.)

iii. FORMAT (Example):

1. SCOPE

1.1 This specification covers the following airplane:

Service model designation.....

Designer's name and model designation.....

Number and places for crew.....

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Number and kind of engines.....

1.1.1 The mission of this aircraft is..... (Copy mission as given in the applicable design requirements document.)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications with issue dates as listed below form a part of this specification:

2.1.1 *(Applicable to U. S. Air Force (USAF).)*

SPECIFICATIONS

MILITARY

MIL–C–5011.....dated.....Charts, Standard Aircraft Characteristics and Performance Piloted Aircraft

MIL–L–6880.....dated.....Lubrication of Aircraft General Specification for

MIL–I–8500.....dated.....Interchangeability and Replaceability, Physical, of Component Parts for Aircraft (Including Guided Missiles)

MIL–A–8806.....dated.....Acoustical Noise Level in Aircraft, General Specification for

U. S. AIR FORCE

Requirements Document.....dated.....

MIL–A–8730.....dated.....Aircraft, General Specification for

MIL–S–5700.....dated.....Structural Criteria, Piloted Airplanes

MIL–F–8785.....dated.....Flying Qualities of Piloted Airplanes

MIL–F–25352.....dated.....Flutter, Divergence, and Reversal in Aircraft, Prevention of

CONTRACTOR PREPARED

(List here the contractor prepared specifications which describe the various equipments and subsystems of the aircraft.)

STANDARDS

MIL–STD–130.....dated.....Identification Marking of U. S. Military Property

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MIL–STD–254.....dated.....Weight and Balance Data Reporting Forms for Aircraft

(*Note:* In addition, list here only those additional standards not cited in 2.1 of other documents and which apply to the aircraft.)

DRAWINGS

(List here those drawings applicable to the item covered by the detail specification.)

PUBLICATIONS

AIR RESEARCH AND DEVELOPMENT COMMAND

Circular Proposal or Contract No. .....

ARDC Manual 80–1 Handbook of Instructions for Aircraft Designers.....Edition including revisions up to and including.....
date

AIR FORCE-NAVY AERONAUTICAL BULLETIN

No. 143 — Specifications and Standards; Use of dated.....

U. S. AIR FORCE SPECIFICATION BULLETIN

No. 59.....dated.....Specifications applicable to the Design and Construction of Air Frames

2.1.1 *(Applicable to Navy — Bureau of Aeronautics (BuAer))*

SPECIFICATIONS

MILITARY

NAVAEROO–25–544 — List of Military Specifications and Standards Used by Bureau of Aeronautics dated.....

BUREAU OF AERONAUTICS

Type Specification.....dated.....

SD–24.....dated.....General Specification for Design and Construction of Aircraft Weapon Systems

STANDARDS

NAVAEROO–25–544 — List of Military Specifications and Standards Used by the Bureau of Aeronautics dated.....

DRAWINGS

NAVAEROO–25–543 — List of Aeronautical Standard Drawings Used by the Bureau of Aeronautics dated.....

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PUBLICATIONS

NAVY DEPARTMENT

Aviation Supply Catalog of Navy Material dated.....

(Copies of the above publication may be obtained from the Aviation Supply Office, 700 Robbins Avenue, Philadelphia 11, Pennsylvania.)

AIR FORCE-NAVY AERONAUTICAL BULLETIN

No. 143 — Specifications and Standards; Use of, dated.....

2.1.1 *(Applicable to the U. S. Army)*

SPECIFICATIONS

MILITARY

MIL-P-11747 — Painting, Marking, and Insignia for U. S. Army Aircraft

(*Note:* In addition, list here those specifications cited in Sections 3, 4, and 5 of the detail specification.)

STANDARDS

MIL-STD-210 — Climatic Extremes for Military Equipment

(*Note:* In addition, list here those standards cited in Sections 3, 4, and 5 of the detail specification.)

DRAWINGS

(*Note:* List here those drawings applicable to the item covered by the detail specification.)

PUBLICATIONS

TB-AVN-7 — Painting and Marking of Army Aircraft

AR-705-15 — Operation of Equipment Under Extreme Conditions or Environment

(*Note:* In addition, list here those publications cited in Sections 3, 4, and 5 of the detail specification.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Characteristics.

3.1.1 *(Three-view drawing).* (A reduced three-view-drawing figure, page size the same as a page of the detail specification and numbered "3.1.1, figure 1" shall be included as the next page following 3.1. Dimensions to be shown shall include at least: span of wing, span of tail, overall length, maximum height, height over fuselage, tread, and wheelbase, and show pilot's vision over the nose of the aircraft.)

3.1.2 *(USAF) Aircraft performance.* (The

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following subparagraphs shall contain performance data as indicated. These data shall be prepared according to the rules and definitions of Specification MIL–C–5011, except that rules and definitions contained in the design requirements document shall govern where they conflict with Specification MIL–C–5011.)

3.1.2.1 *(USAF) Designed aircraft performance.* (Items of performance that are specified as "required" in the design requirements document shall be listed herein and compared with the designed performance. These items shall be clearly identified with the appropriate gross weight, engine power, configuration, etc.)

3.1.2.2 *(USAF) Additional performance data.* (Curves summarizing the performance shall be prepared and included herein. The required performance listed in 3.1.2.1 shall be indicated on the curves. Mission profiles shall be presented for the required mission(s) and shall include fuel allowances, pertinent weights and configurations. Additional estimated performance data may be presented to show other performance capabilities if the designer so desires.)

3.1.2.3 The aircraft performance specified herein is based on the engine specification number (engine manufacturer's specification).

3.1.2 *(BuAer) Guaranteed tabulated performance.* Guaranteed performance shall be for the gross weights with the useful loads specified in the detail specification.

3.1.3 *Weights.* (Tables of MIL–STD–254 Group Weight Statement shall be reproduced here, unless equivalent abstracts are specified by the procuring activity.)

3.1.3.1 *(USAF) Alternate loading.* (Principal alternate loading conditions shall be listed, with suitable explanation of the manner in which they differ from the "Design useful load condition", that is, showing increases or decreases in bomb or cargo load, full load, et cetera. Each of these alternate loading conditions shall be given an identifying number such as "I–A", "I–B", II–A", et cetera, to permit ready and definitive reference to each of these conditions. In addition, a "Useful load" sheet (page 4 of AN–9103 (MIL–STD–254) form) shall be prepared, properly labeled as to "condition", for example, "Alternate useful load I–A", and inserted in the detail specification immediately following the "Design useful load" sheet of the AN–9103 form.)

3.1.4 *Center of gravity locations.*

Design Gross Weight, cg location, basic flight condition:
Aft LE of MAC.....percent MAC
(Above, below) LE of MAC.....inch

Design Gross Weight, cg location, take-off configuration:
Aft LE of MAC.....percent MAC
(Above, below) LE of MAC.....inch

Extreme forward position, cg possible in flight regardless of loading at take-off (wheels up, wheels down):
Aft LE of MAC.....percent MAC
(Above, below) LE of MAC.....inch
Gross Weight for this condition.....lb

Extreme rearward position, cg posible in flight regardless of loading at take-off (wheels up, wheels down):

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Aft LE of MAC.....percent MAC
(Above, below) LE of MAC.....inch
Gross Weight for this condition.....lb

3.1.5 *Areas.* (Items may be added, omitted, or modified in this paragraph to properly describe the aircraft. The following information is not to be used for inspection purposes.)

Wing area, total, including ailerons, flaps and ——square feet of fuelage or hull.....——square feet

Wing flap area, aft of hinge line, total.....——square feet

Aileron area, aft of hinge line, but including overhanging external balance forward of hinge, total including.....square feet of tab area .....——square feet

Horizontal tail area, total.....——square feet

Stabilizer, (to elevator hinge), excluding overhanging elevator balance and including—— square feet of fuselage or hull and.....square feet of contained elevator balance.....——square feet

Elevator, aft of hinge, but including——square feet of overhanging external balance forward of hinge and.....square feet of tab area.....——square feet

Vertical tail area, total.....——square feet

Fin, (to rudder hinge), excluding overhanging rudder balance and including.....square feet of contained rudder balance.....——square feet

Rudder, aft of hinge, but including.....square feet of overhanging external balance forward of hinge and.....square feet of tab area.....——square feet

3.1.6 *Dimensions and general data.* (Items may be added, omitted, or modified in this paragraph to properly describe the aircraft. The following information is not to be used for inspection purposes.)

Wings:
Span, maximum .....————feet
Span, wings folded .....————feet

Chord:
At root .....————inches
At construction tip (theoretical extended section at tip) .....————inches
Mean aerodynamic .....————inches

Airfoil section designation and thickness (percent chord) .....————

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At root .....
At construction tip (theoretical extended section at tip) .....

(*Note:* When an airfoil section other than a NACA section is used, the airfoil section shall be compared with the most nearly equivalent NACA section, or data similar to that available for NACA sections shall be presented.)

Incidence (degrees):
At root .....
At construction tip .....

Sweepback at.....percent chord, degrees.....

Dihedral, degrees .....

Aspect Ratio .....

Ailerons:
Span .....
Chord (average percent wing chord) .....
Tab dimensions .....
Distance from plane of symmetry to centroid of aileron area .....
The amount of the aerodynamic and static balance shall be specified, in percent, and the coefficient of dynamic balance given.....

Spoilers:
Span .....
Chord (average percent wing chord) .....
Position of leading edge (average percent wing chord) .....

High lift and drag increasing device:
Type .....
Span, exclusive of cutouts (percent of wing span) .....
Chord (average percent wing chord).....

Speed brakes:
Span .....
Chord .....
Location .....

Tail:
Horizontal:
Span .....
Chord .....
Airfoil section designation and thickness....
Tab dimensions .....
Tab location .....
The amount of the aerodynamic and static balance shall be specified, in percent, and the coefficient of dynamic balance given....

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(*Note:* When an airfoil section other than a NACA section is used, the airfoil section shall be compared with the most nearly equivalent NACA section, or data similar to that available for NACA sections shall be presented.)

Incidence, normal .....
Sweep of leading edge, degrees.....
Dihedral, degrees .....
Adjustment each side of neutral.....
Aspects ratio .....
Vertical:
Airfoil section designation and thickness.....

(*Note:* When an airfoil section other than a NACA section is used, the airfoil section shall be compared with the most nearly equivalent NACA sections, or data similar to that available for NACA sections shall be presented.)

Sweep of leading edge, degrees.....
Aspect ratio .....
Tab dimensions .....
Tab location .....
The amount of the aerodynamic and static balance shall be specified, in percent, and the coefficient of dynamic balance given....

Height over highest fixed part of aircraft (wing, tail, or propeller disk):
Reference line level (reference is to.....
Wings not folded .....
Wings folded .....
Maximum during folding operation.....

Three-point, or on beaching gear, or handling truck:
Wings not folded .....
Wings folded .....
Maximum during folding operation.....

Height over highest part of tail, reference line level .....

Height in hoisting attitude, from top of hoisting sling to lowest part of aircraft (wheels in lowest position for landplanes and amphibians; on beach handling gear where used).....

Length, maximum:
Reference line level .....
Three point, or on beach handling gear or truck, propeller stationary, lower blade vertical .....

Length from hoisting sling to farthest aft part of tail, reference line level, rudder neutral, elevator down .....

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Distance from wing MAC quarter chord point to horizontal tail MAC quarter chord point.....————

Distance from wing MAC quarter chord point to vertical tail MAC quarter chord point.....————

Angle between reference line and wing zero-lift line .....————

Ground angle, degrees .....————

Propeller clearance, (Normal design) loading condition:
Reference line level, landplane or amphibian....————

Propeller diameter (to nearest inch).....————

Wheel size:
Main wheels .....————
Bicycle wheels, front main——rear main.....————
Auxiliary wheel (nose, tail, or other).....————

Tire size:
Main wheels .....————
Bicycle wheels, front main——rear main.....————
Auxiliary wheel (nose, tail, or other).....————

Tread of main wheels.....————

Wheel base .....————

Vertical travel of axle from extended to fully compressed position:
Main wheels .....————
Nose wheel .....————
Tail wheel (in normal trailing position).....————

Distance from main wheel contact point to center of gravity:
Horizontal Distance:
At most forward c. g. —— at —— lbs gross wt.
At most aft c. g. —— at —— lbs gross wt.
Vertical Distance:
At most forward c. g. —— at —— lbs gross wt.
At most aft c. g. —— at —— lbs gross wt.

Location of external stores from the aircraft center line .....————

3.1.7 *Control movement and corresponding control surface movements.* (Items may be added, omitted, or modified in this paragraph to properly describe the aircraft. The following information is not to be used for inspection purposes.)

Control and control surface movements on each side of neutral position for full movement, as limited by stops.

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Rudder——degrees right——degrees left.

Rudder pedals——inches forward——inches aft.

Rudder tab or trim surface——and degree right——degrees left.

Rudder tab of trim surface control——turns for——degrees of tab or trim surface movement.

Elevators——degrees above——degrees below.

Elevator control——inches aft——inches forward.

Elevator tab or trim surface——degrees above——degrees below.

Elevator tab or trim surface control——turns for——degrees of tab or trim surface movement.

Ailerons——degrees above——degrees below.

Aileron control stick——inches right——inches left.

Aileron control wheel——degrees right——degrees left.

Aileron droop (maximum)——degrees.

Aileron tab——degrees above——degrees below.

Aileron tab control——turns for——degrees tab movement.

Stabilizer wheel.....turns for.....degrees of stabilizer movement.

Wing flap (maximum)——degrees movement.

Wing flap control——strokes or turns for——degrees flap movement.

Water rudder——degrees each side.

Speed brake control——degrees for——degrees of speed brake movement.

3.2 General features of design and construction.

3.2.1 *General interior arrangement.* (Includes personnel and major equipment items — may be supplemented with reduced inboard profile drawing and cross sections, in which event, the drawing shall follow the three-view drawing called for in 3.1.1).

3.2.2 *Selection of materials.* (Sample statement): Specifications and standards for all materials, parts, processes, and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, and other applicable design criteria documents of the procuring activity.

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3.2.3 *Workmanship.*

3.2.4 *Production, maintenance, and repair.* (Sample statement) : The design of the aircraft shall be such as to insure ease of production, easy and rapid installation of the powerplant and equipment, and ease of general maintenance. Special attention shall be given to the ease with which the component parts of the structure and installations can be inspected, maintained, and repaired. The fuselage and wings shall be designed to facilitate the removal and replacement of damaged sections. The members that are most liable to damage shall be designed in a manner to preclude long, laborious processes in their repair and replacement. Requirements for special tools shall be kept to a practicable minimum. (The designer shall insert additional information providing pertinent details.)

3.2.5 *Interchangeability and replaceability.* (Sample statement) : Assemblies, components, and parts of all aircraft of the same model which are not peculiar to a particular series or block number designation shall be interchangeable or replaceable in accordance with and to the extent required by Specification MIL–I–8500 and the contractor's approved interchangeability-replaceable working list.

3.2.6 *Finish.*

3.2.6.1 *(USAF).* The finish for the aircraft and parts shall be in accordance with

Specification ———————————————
(Mfr's Finish Spec.)

dated ———————————.

3.2.7 *Identification and marking.* (Sample statement) : The aircraft and its components shall be identified and otherwise marked in accordance with the provisions of Standard MIL–STD–129.

3.2.8 *Extreme temperature operation.* (Sample statement) : The aircraft as a whole, including aeronautical equipment, shall be so constructed that it will function satisfactorily in any or all temperature conditions that will be encountered. A ground temperature range of —65° +160°F. has been established as the range at which the aircraft shall operate.

3.2.9 *Climatic requirements.* (Sample statement) : The aircraft and its equipment shall not be adversely affected by other climatic conditions incident to the above temperature range, and shall be capable of transfer from one climate to another without penalty of extensive modification and adjustment.

3.2.10 *Lubrication.* (Sample statement) : Lubrication of the aircraft shall conform to the requirements of Specification MIL–L–6880.

3.2.11 *Equipment and furnishing installation.* The equipment and furnishings specified in appendix I-A, I-B, and I-C of this specification and as described in other portions of this specification shall be installed in the quantity and under the applicable conditions set forth.

3.2.12 *(USAF) crew.* (List provisions for and duties of crew as specified in the applicable design requirements document.) (Example: The crew shall consist of the following:

1 Pilot.

1 Copilot.

1 Radio Operator-navigator.

1 Radio Operator-gunner.

1 Tail gunner.

3.2.13 *Noise and vibration requirements.* (Sample statement) : The aircraft and its equipment shall function normally in all extremes of noise and vibration that will be encountered.

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3.2.13.1 *Vibration.* (Sample statement): The design of the airplane shall be such that the probability of disastrous fatigue failure resulting from vibrations (including those produced by noise or other oscillatory air flows) during the service life (insert flight hours of life) of the airplane will not exceed ————. In those types of structure where redundancy, slow rate of crack growth and inspection enable repair in time to avoid disastrous fatigue failure or reduction in strength, the design of the airplane shall be such that the repair rate (i.e., total man-hours of repair time divided by the life of the airplane in flight hours) in normal service operation does not exceed ————.

3.2.13.2 *Crew comfort.*

3.2.13.2.1 *Acoustical noise.* (Sample statement): Acoustical noise shall not exceed the limits stated in Specification MIL-A-8806 and shall be measured as required therein.

3.2.13.2.2 *Vibration.* (Sample statement): The vibration of crew seats, rudder pedals, heel troughs, control columns, work tables, and primary structural members in all occupied areas during unaccelerated flight from minimum to maximum cruising speed shall not exceed ———— acceleration at frequencies below ——— cps, ——— inches double amplitude at frequencies between ——— cps and ——— cps.

3.2.14 *Aircraft sustained operation.* (Sample statement): The aircraft shall be capable of performing missions within the performance envelope on a sustained basis. That is, the aircraft is expected to be available for operational missions approximately ——— percent of the time.

3.3 Aerodynamics. (The detailed requirements pertaining to aerodynamic characteristics of the aircraft shall be included as subparagraphs of this paragraph.)

3.3.1 *General.* (Give general overall characteristics in this paragraph. The designer shall use this section to describe or specify any aerodynamic features or devices not described in detail elsewhere in this specification. Examples are: Smoothness of exterior joints, fairness of surfaces and trueness of contours, cleanness of intersections, sealing against air leakage which may cause drag, and similar items.)

3.3.2 *Stability and control.*

3.3.2.1 *(Sample statement):* Stability and control characteristics shall be in accordance with the requirements of Specification MIL-F-8785.

3.3.2.2 *Center of gravity limits.* (See 3.1.4) (The designer may insert any additional information pertinent to center of gravity travel and its effect on stability and control.)

3.3.3 *(USAF) Flutter characteristics.* (Sample statement): The aircraft and all its components shall be free from divergence, flutter, buzz, or other aeroelastic instability throughout the entire range of permissible speeds, altitudes, maneuvers, aerodynamic and acceleration loading conditions, and weight conditions as determined by methods and criteria in Specification MIL-F-25352.

3.4 (USAF) Structural design criteria. (Sample statement): Structural design of this aircraft is in conformance with the criteria specified in Specification MIL-S-5700. Structural parameters are listed herein.

3.4.1 *Limit flight load factors.* (Note: The information to be supplied below shall be based on certain typical loading conditions specified in the design requirements documents or the applicable strength specification. Any information at variance with the design requirements document or the strength specification shall be covered by acceptable deviations and listed in appendix II. The design weights listed in the following

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subparagraphs are defined in Specification MIL–S–5700.)

3.4.1.1 Minimum Flying Gross Weight ——— lb.
Maneuver
Positive ——— Negative ———
Gust
Positive ——— Negative ———

3.4.1.2 Basic Flight Design Gross Weight ——— lb.
Maneuver
Positive ——— Negative ———
Gust
Positive ——— Negative ———

3.4.1.3 Maximum Design Gross Weight ——— lb.
Maneuver
Positive ——— Negative ———
Gust
Positive ——— Negative ———

3.4.1.4 List any other alternate mission take-off gross weights desired, with associated load factors.

3.4.2 Landing Load Factors.

3.4.2.1 Basic Landplane Gross Weight ——— lbs.
Sinking Speed ——— feet per second.

3.4.2.2 Maximum Landing Gross Weight ——— lbs.
Sinking Speed ——— feet per second.

3.4.3 *Design speeds.* See Figure ———. (The design speeds for the aircraft shall be presented in chart form. The chart shall include the variations of the equivalent level flight high speed with both normal and military rated power, equivalent level flight limited speed (if applicable) and the maximum equivalent design speed plotted versus altitude. Constant Mach number lines shall be superimposed upon the chart along with a plot of stall speed (Vs) for one "g" level flight and any other speeds to which the aircraft will be restricted because of gross weight, temperature, etc).

3.4.4 *Center of gravity limits.* See Figure ———: (The center of gravity envelope shall be presented in chart form plotted versus gross weight. The chart shall show the operational limits and the limits to be used for structural design.)

3.4.5 *Structural design service life.* The structural design will demonstrate freedom from failure during the service life for ——— of flying hours and ——— of landings.

3.4.6 *Ditching criteria.* (Any detailed requirements pertaining to ditching criteria shall be included as subparagraphs of this paragraph.)

3.4 (BuAer) Structural design criteria.

3.4.1 *Strength requirements.* (This paragraph shall summarize the essential strength characteristics of the aircraft, and shall contain the following statement: "These data are shown for information only and apply to the aircraft as conceived as of the date of the detail specification. The contractual obligation of the contractor under this specification shall be as set forth in the succeeding paragraphs."
Example:

Basic flight design gross weight ——— lb
Basic landing design gross weight ——— lb.
Basic catapulating design gross weight ——— lb.
Basic landplane take-off design gross weight ——— lb.
Rough water operation take-off design gross weight ——— lb.
Sheltered water operation design gross weight ——— lb.
Positive limit load factor at basic flight design gross weight ——— lb.
Design sinking speed, ——— fps.)

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3.4.1.1 *Detail strength requirements.* (The detail strength requirements shall be generally in the form of modifications and amplification of the requirements in Specifications MIL–A–8629 or MIL–S–5700 series as shown in the detail specification.)

3.5 Wing group.

3.5.1 *Description and components.* (Give general description of the wing, including type of wing, number and arrangement of components, e.g., center section, outer panels, tips.)

3.5.2 *Construction.* (This paragraph shall include a description of type of construction and materials. Such information shall be given to identify positively the material, either by reference to the procuring activity's specification or description of type and properties. Include general description of load-carrying structure, spars, beams, stringers, ribs, leading edge structure, skin, auxiliary beams, joints, et cetera, and (briefly) the basic general assumptions made relative to the functions of the various members in carrying bending, torsion, and shear loads. It is desired that assumptions pertaining to the stress path and load-carrying members at discontinuities be explained briefly.)

3.5.3 Ailerons: (Include a description of the ailerons and tabs, giving type, construction, type of aerodynamic and dynamic balance, and material used.)

3.5.4 *Lift and drag devices.*

3.5.5 *Speed brakes (if on wing).*

3.6 Tail group.

3.6.1 *Description and components.* (A general description shall be given, including what comprises tail group.)

3.6.2 *Stabilizer.* (A description shall be given, including construction and materials.)

3.6.3 *Elevator.* (Include a description of the elevator and tabs, giving type, construction, type of aerodynamic and dynamic balance, and materials used.)

3.6.4 *Fin.* (A description shall be given, including construction, materials, et cetera.)

3.6.5 *Rudder.* (Include a description of the rudder and tabs, giving type, construction, type of aerodynamic and dynamic balance, and materials used.)

3.7 Body group.

3.7.1 *Fuselage.*

3.7.1.1 *Description.* (A general description of the fuselage shall be given, including crew locations; location of principal equipment, compartments, pressurized section, et cetera.)

3.7.1.2 *Construction.* The type, joints, materials, et cetera, and (briefly) the basic general assumptions made as to the structural functions of the members shall be specified.)

3.7.1.3 *Crew station subsystems.* (Describe each station in separately numbered and titled subparagraphs covering such items as, but not limited to: Personnel furnishings, accommodations, conveniences; instrument and switch panel locations and configurations (include sketches); lighting and visibility (internal-external); survival and crash restraining provisions; escape provisions; oxygen; controls-personnel and equipment location arrangements, range of motions, interferences, and clearances; exits, including location and size; access path from entrance to station; soundproofing; et cetera.)

3.7.1.3.1 *Oxygen system.* (The entire oxygen system shall be described in detail to include such items as — type (liquid or gaseous), type and number of supply containers, regulators (type and number), walk-around

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provisions (if applicable) filling provisions, gaging provisions, designed supply capabilities, and any other data required to completely describe the oxygen system. All applicable specifications which define the system shall be listed.)

3.7.1.4 *Cargo compartments.* (The size, flooring, floor loads, structural provisions for tiedown, soundproofing, et cetera, shall be given.)

3.7.1.5 *Equipment compartments.*

3.7.1.6 *Speed brakes (if on fuselage).*

3.7.2 *Hull.*

3.7.2.1 *Description and arrangement.*

3.7.2.2 *Design and construction.*

3.7.2.3 *Crew and personnel stations.*

3.7.2.4 *Cargo compartments.*

3.7.2.5 *Equipment compartments.*

3.8 **Alighting gear.** (Include a description of the wheel, or equivalent type alighting gear in applicable subparagraphs.)

3.8.1 *General description and components.* (A general description and arrangement shall be given.)

3.8.2 *Main landing gear.*

3.8.2.1 *Description.* (A general description of suspension and listing of the main components, including structure, materials, et cetera, shall be given.)

3.8.2.2 *Wheels, brakes, and brake control systems.* (Give size, types, et cetera.)

3.8.2.3 *Casing and tubes.* (Sizes, types, et cetera).

3.8.2.4 *Shock absorbers.* (Give types, et cetera).

3.8.2.5 *Retracting, extending, and locking systems.* (Motions, time of operation, power means, linkages, locks, emergency provisions, et cetera, shall be described.)

3.8.2.6 *Doors and fairings.* (Give materials, operation, function, et cetera.)

3.8.2 ( ) *Inspection and maintenance.* (Describe provisions for lubrication, servicing, removal, et cetera, Parentheses in paragraph number are used to designate this paragraph as the last subparagraph under 3.8.2. This paragraph shall be numbered accordingly.)

3.8.3 *Auxiliary landing gear (tail wheel).* (See 3.8.2 and subparagraphs—similar kind of information shall be given.)

3.8.3.1 *Description.*

3.8.3.2 *Wheels.*

3.8.3.3 *Casings and tubes.*

3.8.3.4 *Shock absorbers.*

3.8.3.5 *Retracting, extending, and locking systems.*

3.8.3.6 *Doors and fairings.*

3.8.3.7 *Steering control.*

3.8.3 ( ) *Inspection and maintenance.* (Parentheses in paragraph numbers are used to designate this paragraph as the last subparagraph under 3.8.3. This paragraph shall be numbered accordingly.)

3.8.4 *Auxiliary landing gear (nose wheel).* (See 3.8.2 and subparagraphs—similar kind of information shall be given.)

3.8.4.1 *Description.*

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3.8.4.2 *Wheels.*

3.8.4.3 *Casings and tubes.*

3.8.4.4 *Shock absorbers.*

3.8.4.5 *Retracting, extending, and locking systems.*

3.8.4.6 *Doors and fairings.*

3.8.4.7 *Steering control.*

3.8.4 *( ) Inspection and maintenance.* (Parentheses in paragraph number are used to designate this paragraph as the last subparagraph under 3.8.4. This paragraph shall be numbered accordingly.)

3.8.5 *Auxiliary landing gear (other).* (See 3.8.2 and provide appropriate subparagraphs.)

3.8.5.1 *Description.*

3.8.6 *Flotation characteristics.*

3.9 Alighting gear (water type).

3.9.1 *General description and components.*

3.9.2 *Main floats.*

3.9.3 *Auxiliary floats.*

3.10 Flight control system.

3.10.1 *Primary flight control systems.* (Give general descritpion including at least type; routing through airplane; vulnerability, including duplication of circuits; cable and component sparing; stops; adjustments; locks; et cetera.)

3.10.1.1 *Flight station controls.*

3.10.1.2 *Lateral system.* (Describe the system including boost or power, actuators, mechanisms, accessibility, et cetera.)

3.10.1.3 *Directional system.* (Describe the system including boost or power, actuators, mechanisms, accessibility, et cetera.)

3.10.1.4 *Longitudinal system.* (Describe the system including boost or power, actuators, mechanisms, accessibility, et cetera.)

3.10.2 *Secondary flight control systems.*

3.10.2.1 *Lift and drag increasing device(s) systems.*

3.10.2.2 *Speed brakes.*

3.10.3 *Trim (or tab) control systems.* (Give general description, including longitudinal, directional, and lateral trim control systems.)

3.10.4 *Automatic flight control system.* (Provide a general description of the system including the functional tie-ins with other systems such as bombing, fire control, navigation, et cetera. Reference shall be made to the procurement specification which contains the detail functional and performance requirements for the system.)

3.11 Engine section or nacelle group.

3.11.1 *Description and components.*

3.11.2 *Construction.*

3.11.3 *Engine mounts.*

3.11.4 *Vibration isolators.*

3.11.5 *Firewalls.* (Give location, materials, method of sealing, et cetera.)

3.11.6 *Cowling and cowl flaps.* (Type, method of actuation, et cetera.)

3.11 *( ) Access for inspection and maintenance.* (Parentheses in paragraph number are used to designate this paragraph as the last subparagraph under 3.11. This paragraph shall be numbered accordingly.)

3.12 Propulsion subsystem.

3.12.1 *General description and components.*

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(Discuss generally the number, types, and arrangement of main and auxiliary propulsion units. Describe the propulsion subsystem and include such parts of the subsystem as the power control subsystem, fuel supply and fuel injection subsystem, lubrication subsystem, starting subsystem, air-inlet configuration and controls subsystem, thrust augmentation or reversal subsystem et cetera.)

3.12.2 *Main propulsion unit(s).* (List the type of propulsion unit and its specification, and describe the design features which allow for ease of installation and removal, and access for inspection and maintenance.)

3.12.3 *Auxiliary propulsion unit(s).* (List the type of propulsion unit and its specification, and describe the design features which allow for ease of installation and removal and access for inspection and maintenance.)

3.13 Secondary power and distribution subsystems.

3.13.1 *Electrical power generation and distribution subsystem.* (Describe the type and features of the electrical power generation and distribution subsystem and list the electrical system loads. List the overall specification for the electrical subsystem.)

3.13.2 *Hydraulic power generation and distribution subsystem.* (Describe the general design features of the subsystem, list the hydraulically actuated services, and list the overall specification for the subsystem. State the estimated maximum fluid temperature and the estimated total pump capacity.)

3.13.3 *Pneumatic power generation and distribution subsystem.* (Describe the general design features of the subsystem, list the pneumatically actuated services, and list the overall specification for the subsystem.)

3.14 Utilities and equipment subsystems. (Provide a general description of each of the following subsystems and list the specifications which describes the subsystems in more detail: Air conditioning, anti-icing, defogging, moisture control; pressurization; fire radiation hazard detection, prevention, and protection.)

3.15 Mission and air traffic control subsystems. (Provide a general description of each of the following subsystems and list the specification which describes the subsystem; communications, navigation, and identification.)

3.16 Reconnaissance subsystems. (Provide a general description of each of the following subsystems and list the specification which describes the subsystem; Terrain reconnaissance; air target reconnaissance; atmospheric reconnaissance; et cetera.)

3.17 Fire-power control subsystems. (Provide a general description of each of the following subsystems and list the specification which describes the subsystem; bombing-navigation; optical-bombing, guidance; fire control; et cetera.)

3.18 Armament subsystems. (Provide a general description of each of the following subsystems and list the specification which describes the subsystems; guns; missile-launching; torpedo-launching; atomic and thermonuclear weapons; molecular bomb munitions; et cetera.)

3.19 Cargo and transport subsystems. (Provide a general description of each of the following subsystems and list the specification which describes the subsystem. Include the size, flooring, floor loads, structural provision for tiedown et cetera; personnel handling; cargo handling; and cargo servicing.)

3.20 Countering subsystems. (Provide a general description of each of the following subsystems in a separate subparagraph and list the specification which describes the subsystem: Electronic Countermeasures; Jamming; Deception; Chaff Dispensing; Infra-

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Red Countermeasures; et cetera.)

3.21 Ground handling and servicing provisions. (Describe the provisions and features for such functions as: towing, packing, mooring, hoisting, leveling, servicing, wing securing (carrier aircraft), catapulting (carrier aircraft) barrier crash, arresting hook (carrier aircraft), et cetera.)

3.22 Aerial resupply subsystems.

3.23 Air rescue subsystems.

3.24 Range extension subsystems.

3.25 Air weather subsystems.

3.26 Preflight readiness checkout provisions. (Describe the provisions and features which have been incorporated to permit a rapid and thorough readiness checkout of the aircraft and its installed systems and equipments.)

4. QUALITY ASSURANCE PROVISIONS

4.1 (USAF) Unless otherwise specified by the procuring activity, the method of inspection and tests shall be in accordance with Specification MIL–A–8730 dated————, and applicable specifications for specific subsystems.

4.1 (BuAer) The method of inspection and tests shall be as specified by the Bureau of Aeronautics.

5. PREPARATION FOR DELIVERY

5.1 The aircraft shall be prepared for delivery as specified by the procuring activity.

6. NOTES

6.1 Intended use. (Sample statement): This specification is intended to be used as the contract document which describes the aircraft supplied under terms of the contract.

Note: **Include as additional paragraphs under this section definitions or other notes which are pertinent.**

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APPENDIX I

Equipment lists (Items in Appendixes I–A, I–B, I–C to be segregated under subsystem or group headings, e.g., Propulsion, Instruments, et cetera.) (Unless otherwise specified, items under each heading should be presented in the following order:

Item No. *Quantity* *Description* *Identification* *Unit Weight (lb)*

Appendixes I–A and I–B shall consist of the Government-Furnished Aircraft Equipment List appropriately titled.

APPENDIX I–A

Government-Furnished Aircraft Equipment; Contractor-Installed
(Example)

Propulsion System (Weight Empty)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	4	Engine, Pratt & Whitney	R–4360 PW Spec A–7051	3,513.00
.....	.....	.....	.....	.....
.....	.....	.....	.....	.....

Instruments (Weight Empty)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	7	Altimeter	AN5760–2A	1.40
.....	.....	.....	.....	.....
.....	.....	.....	.....	.....

APPENDIX I–B

Government-Furnished Aircraft Equipment, Government Installed
(Example)

Propulsion System (Weight Empty)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	.....	.....	.....	.....
2	.....	.....	.....	.....

Instruments (Weight Empty)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	1	Compass, astro	AN5738–1	2.12
2	1	Standard, astro-compass	AN5738–2	1.00
.....	.....	.....	.....	.....
.....	.....	.....	.....	.....

GPO 802693–2

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Armament (Design Useful Load)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	6,000	Rounds 50-cal ammunition with links	.....	0.30
.....	.....	.....	.....	.....
.....	.....	.....	.....	.....

Armament (Alternate Useful Load)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	7,000	Rounds 50-cal ammunition with links	.....	0.30
.....	.....	.....	.....	.....
.....	.....	.....	.....	.....

APPENDIX I–C

Contractor-Furnished, Contractor-Installed (Example)

Propulsion System (Weight Empty)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	16	Valve Assy, fuel control	9–10776	1.11
.....	.....	.....	.....	.....
.....	.....	.....	.....	.....

Instruments (Weight Empty)

Item No.	*Quantity*	*Description*	*Identification*	*Unit Weight (lb)*
1	1	Indicator flap position	3–39263	1.25
2	5	Amplifier, fuel level	15–19423	1.10
.....	.....	.....	.....	.....
.....	.....	.....	.....	.....

APPENDIX II

Deviations

Note all deviations from Government specifications, from procuring activity designers handbook requirements and related documents, and from military design requirements. Note deviations common to models covered in the detail specification, then add addendum paragraphs cover-

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ing these deviations which are peculiar to any one or more alternates. Deviation items shall be numbered consecutively, shall reference the paragraph in the detail specification where the deviation occurs, shall reference the document and the requirement in the document against which deviation is taken, shall state the deviation, and give reasons for the deviation. Deviation numbers shall remain the same for all subsequent detail specifications of a particular aircraft model. If at any time a deviation has been deleted after the first issue of the detail specification, the number will remain, with a note that the deviation has been deleted.

(Example)

Deviation	*Paragraph*	*Oleo strut air pressure*
1	3.8.2.4	

Requirement:

Deviation:

Reason for change and remarks:

APPENDIX III (for USAF, only)

(See paragraph 3.9 *Changes to aircraft Detail Specification*)

For ECP No. .....Date.....TMS.....
File Opposite Page No. .....of Specification.....
Contractual authority.....Contract.....Effectively.....
.....
.....
.....
.....
Effect of change: (List by specification paragraph number.)

Note: Include verbatim statement of revision to the aircraft detail specification which shall call out the specification paragraph reference, and be one of the following alternate forms:

(a) Add
(b) Delete
(c) Change from.....to.....

APPENDIX IV

Optional arrangements (for USAF proposals only)

APPENDIX IV-A, IV-B, IV-C, et cetera:

(Assign a definite manufacturer's model number for each alternate arrangement, the model number to consist of the basic model number with a suitable suffix designation. A proposal shall be considered as an alternate only when there is a change in a major component of the basic design, e.g., use of a different type or model of engine, different fuselage or crew arrangement, a change in wing area, and entirely different type of empennage or alighting gear assembly, alternat armament installation, et cetera.

Note: The form of each appendix shall correspond to the form prescribed for the detail specification covering the basic proposal. Only those paragraphs that are affected

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by the change, or changes, from the basic design are to be rewritten and reproduced in the Appendixes. Rewritten paragraphs shall bear the same number as the corresponding paragraph for the basic design. Each rewritten paragraph so numbered is considered as automatically superseding the basic paragraph, and a statement to this effect shall be included. Each appendix shall contain a list of drawings and reports specifically applicable to the alternate proposal covered in that appendix. The list shall give drawing and report number titles, and a reference to the number of the corresponding drawing or report of the basic design.

APPENDIX V

Summary of optional arrangements (for USAF proposals only)

(Present a summary of the basic design and all alternates, in tabular form showing aircraft model numbers, general arrangement and inboard profile drawing numbers, complete performance figures, design gross weight, weight empty, type and model of engines with ratings and any other major differences such as wing area, alighting gear type, crew arrangement, empennage design, design useful load fuel and fuel capacities, et cetera.)

4. PREPARATION FOR DELIVERY

4.1 The detail specification shall be prepared for delivery as specified by the procuring activity.

5. NOTES

5.1 Intended use. The information contained in this specification is intended to be used as the basis for the preparation of detail specifications covering aircraft.

5.2 Additional information.

5.2.1 *Specification for aircraft for U. S. Army.* Requirements of the detail specification as it relates to U. S. Army aircraft shall be in accordance with the normal procedures of the procuring activity.

5.3 Definitions.

5.3.1 The Handbook of Instructions for Aircraft Designers, ARDC Manual 80-1, and the General Specification SD-24 contain definitions pertaining to aircraft of the USAF and the U. S. Navy, respectively. Any other notes which the bidder or manufacturer desires to incorporate should be included in section 6 of the aircraft detail specification.

5.3.2 Where the phrases "complete provision for", "structural provision for", and "space provision for" are used, the intent is defined as follows.

5.3.2.1 *Complete provisions for.* "Complete provisions for" a specific item of equipment or assembly or installation shall mean that all supports, brackets, tubes and fittings, electrical wiring, hydraulic lines, et cetera,

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have been installed and adequate weight and space allocated in order that the equipment can be installed without alteration to the specific equipment or the airplane, and that no additional parts are required for installation, other than the item itself. Standard stock items such as nuts, bolts, cotter pins, et cetera, need not be furnished.

5.3.2.2 *Structural provisions for.* "Structural provisions for" a specific installation shall mean that the primary structure will be structurally adequate for the installation, but that brackets, bolt holes, electrical wiring, hydraulic lines, et cetera, will not be required. Structural provisions also include weight of the equipment involved as an element of alternate weight.

5.3.2.3 *Space provisions for.* "Space provisions for" a specific installation shall mean that space only shall be allocated for the installation, and that brackets, bolt holes, electrical wiring, hydraulic lines, et cetera, will not be required. "Space provision for" does not simply imply that adequate attacking structure is provided, unless otherwise stated.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Transportation Corps
Navy—Bureau of Aeronautics
Air Force

Preparing activity:
Air Force

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